Vertex Model Master Derivations: Hessian and Shear Modulus

Gemini CLI | Expert Computational Biophysicist

March 26, 2026

1 Introduction

This document provides the definitive mathematical reconciliation between standard literature (Merkel & Manning 2018) and the analytical derivations in `vm_human_calculations`. It serves as the canonical reference for implementing the Hessian matrix ($\mathcal{H}$) and the macroscopic shear modulus (g) under Lees-Edwards Boundary Conditions (LEBC).

2 Definitions & State Variables

The energy functional E for a tissue of N_c cells is defined as:

$$E = \sum_{\alpha=1}^{N_c} \left[\frac{K_A}{2}(A_\alpha - A_0)^2 + \frac{K_P}{2}(P_\alpha - P_0)^2 \right] \tag{1}$$

2.1 Conjugate Variables (Stress Coefficients)

To facilitate higher-order derivatives, we define the following coefficients:

- **Cell Pressure:** $\Pi_\alpha = -\frac{\partial E}{\partial A_\alpha} = -K_A(A_\alpha - A_0)$
- **Edge Tension:** $T_\alpha = \frac{\partial E}{\partial P_\alpha} = K_P(P_\alpha - P_0)$

2.2 Physical Distance and LEBC Unwrapping

Under simple shear strain γ, the physical distance $\Delta\mathbf{r}_{ij}$ must be unwrapped to account for the periodic image shift. For a box of dimensions (L_x, L_y):

$$n_y = \text{round}\left(\frac{y_j - y_i}{L_y}\right) \tag{2}$$

$$\Delta y = (y_j - y_i) - n_y L_y \tag{3}$$

$$\Delta x = (x_j - x_i) - n_y(\gamma L_y) - L_x \cdot \text{round}\left(\frac{(x_j - x_i) - n_y(\gamma L_y)}{L_x}\right) \tag{4}$$

3 First Derivatives (Forces)

The force on vertex i is $\mathbf{f}_i = -\frac{\partial E}{\partial \mathbf{r}_i}$.

$$\frac{\partial E}{\partial \mathbf{r}_i} = \sum_{\alpha \in \text{cells}(i)} [-\Pi_\alpha \boldsymbol{\nabla}_i A_\alpha + T_\alpha \boldsymbol{\nabla}_i P_\alpha] \tag{5}$$

3.1 Geometric Gradients

- **Area Gradient:** $\boldsymbol{\nabla}_i A_\alpha = \frac{1}{2}\hat{\mathbf{z}} \times (\mathbf{r}_{i+1} - \mathbf{r}_{i-1}) = \frac{1}{2}(y_{i+1} - y_{i-1}, x_{i-1} - x_{i+1})^T$.
- **Perimeter Gradient:** $\boldsymbol{\nabla}_i P_\alpha = \sum_{j \in \text{neigh}(i)} \hat{\mathbf{t}}_{ij}$, where $\hat{\mathbf{t}}_{ij} = \frac{\mathbf{r}_i - \mathbf{r}_j}{|\mathbf{r}_i - \mathbf{r}_j|}$ is the unit vector pointing *away* from neighbor j toward vertex i.

4 The Hessian Matrix ($\mathcal{H}$)

The Hessian $H_{ij}^{\mu\nu} = \frac{\partial^2 E}{\partial r_i^\mu \partial r_j^\nu}$ is rigorously separated into **Modulus (Stiffness)** and **Stress (Geometric)** components.

$$\mathcal{H}_{ij} = \mathcal{H}_{ij}^{\text{Mod}} + \mathcal{H}_{ij}^{\text{Stress}} \tag{6}$$

4.1 Modulus Terms

$$\mathcal{H}_{ij}^{\text{Mod}} = \sum_\alpha [K_A(\boldsymbol{\nabla}_i A_\alpha \otimes \boldsymbol{\nabla}_j A_\alpha) + K_P(\boldsymbol{\nabla}_i P_\alpha \otimes \boldsymbol{\nabla}_j P_\alpha)] \tag{7}$$

4.2 Stress Terms

$$\mathcal{H}_{ij}^{\text{Stress}} = \sum_\alpha [-\Pi_\alpha \boldsymbol{\nabla}_i \boldsymbol{\nabla}_j A_\alpha + T_\alpha \boldsymbol{\nabla}_i \boldsymbol{\nabla}_j P_\alpha] \tag{8}$$

- **Area Stress:** For $j = i+1$, $\boldsymbol{\nabla}_i \boldsymbol{\nabla}_{i+1} A_\alpha = \frac{1}{2}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.
- **Perimeter Stress:** For connected vertices $i \neq j$, $\boldsymbol{\nabla}_i \boldsymbol{\nabla}_j P_\alpha = -\frac{1}{l_{ij}}(\mathbf{I} - \hat{\mathbf{t}}_{ij} \otimes \hat{\mathbf{t}}_{ij})$.

5 Strain Derivatives under LEBC

Calculating the shear modulus requires first and second total derivatives of the energy with respect to the shear strain γ. In a periodic box (L_x, L_y), the Lees-Edwards boundary condition (LEBC) introduces specific γ-dependencies through the unwrapping shift.

5.1 Area Strain Derivatives

Contrary to standard affine transformations where $\det \mathbf{\Lambda} = 1$ implies area invariance, the *unwrapped* area of a cell crossing the y-boundary under LEBC has a non-zero first derivative:

$$\frac{\partial A_\alpha}{\partial \gamma} = L_y \left[\bar{y}_e^{(L\alpha)} - \bar{y}_e^{(R\alpha)} \right] \tag{9}$$

where $\bar{y}_e^{(L\alpha)}$ and $\bar{y}_e^{(R\alpha)}$ are the average y-coordinates of the vertices on the left and right edges that cross the periodic boundary. The second derivative is zero: $\frac{\partial^2 A_\alpha}{\partial \gamma^2} = 0$.

5.2 Perimeter Strain Derivatives

The first derivative represents the projection of the horizontal shear displacement onto the edge axis:

$$\frac{\partial P_\alpha}{\partial \gamma} = \sum_{e \in \text{crossing}(\alpha)} \xi(\mathbf{l}_e) L_y \frac{\mathbf{l}_e \cdot \hat{\mathbf{x}}}{l_e} = \sum_{e \in \text{crossing}(\alpha)} \xi(\mathbf{l}_e) L_y \cos\theta_e \tag{10}$$

where $\xi(\mathbf{l}_e) = \pm 1$ accounts for the orientation of the crossing edge relative to the boundary traversal (CCW).

The second derivative captures the curvature of the edge length under shear:

$$\frac{\partial^2 P_\alpha}{\partial \gamma^2} = \sum_{e \in \text{crossing}(\alpha)} L_y^2 \frac{\sin^2 \theta_e}{l_e} \tag{11}$$

where θ_e is the angle between the crossing edge and the x-axis.

6 The Affine Force Vector ($\mathbf{\Xi}$)

The vector $\mathbf{\Xi}_i^\mu = \frac{\partial^2 E}{\partial r_i^\mu \partial \gamma}$ represents the forces generated on vertices by an infinitesimal affine box shear. It involves mixed derivatives:

$$\mathbf{\Xi}_i^\mu = \sum_{\alpha \in \text{cells}(i)} \left[K_A \frac{\partial A_\alpha}{\partial \gamma} \frac{\partial A_\alpha}{\partial r_i^\mu} + K_P \frac{\partial P_\alpha}{\partial \gamma} \frac{\partial P_\alpha}{\partial r_i^\mu} - \Pi_\alpha \frac{\partial^2 A_\alpha}{\partial \gamma \partial r_i^\mu} + T_\alpha \frac{\partial^2 P_\alpha}{\partial \gamma \partial r_i^\mu} \right] \tag{12}$$

6.1 Mixed Geometric Derivatives

From `vm_human_calculations`:

- **Area-Strain Mixed:** $\frac{\partial^2 A_\alpha}{\partial \gamma \partial r_i^\mu} = \frac{L_y}{2} \delta_{\mu,y} (\delta_{i \in R\alpha} - \delta_{i \in L\alpha})$.
- **Perimeter-Strain Mixed:** $\frac{\partial^2 P_\alpha}{\partial \gamma \partial r_i^\mu} = \delta_{i \in \alpha} L_y \sum_{e \in \text{crossing}} \delta_{i \in e} \xi(\mathbf{l}_e) \Theta(\mathbf{r}_i) \frac{l_e^2 \delta_{x\mu} - (\mathbf{l}_e \cdot \hat{\mathbf{x}})(\mathbf{l}_e \cdot \hat{\mu})}{l_e^3}$.

7 Linear Response Derivation (Merkel & Manning 2018)

7.1 The Macroscopic Shear Modulus (g)

We define g as the second total derivative of the energy per cell. Following the convention $u = 2g\gamma^2$ from the human notes:

$$g = \frac{1}{4N_c}\frac{d^2E^{\min}}{d\gamma^2} = \frac{1}{4N_c}\left[\frac{\partial^2 E}{\partial\gamma^2} - \boldsymbol{\Xi}^T\mathcal{H}^{-1}\boldsymbol{\Xi}\right] \tag{13}$$

where $\frac{\partial^2 E}{\partial\gamma^2}$ is the affine stiffness and $\boldsymbol{\Xi}^T\mathcal{H}^{-1}\boldsymbol{\Xi}$ is the non-affine relaxation.

7.2 Affine Stiffness ($\mathcal{K}$)

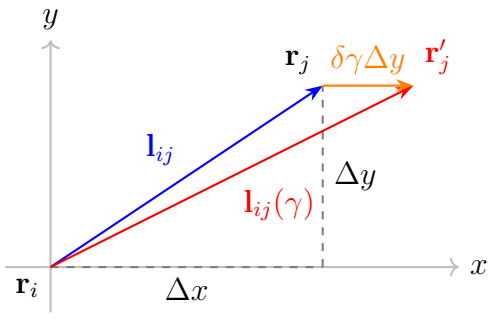
$$\mathcal{K} = \frac{\partial^2 E}{\partial\gamma^2} = \sum_\alpha \left[K_A\left(\frac{\partial A_\alpha}{\partial\gamma}\right)^2 + K_P\left(\frac{\partial P_\alpha}{\partial\gamma}\right)^2 + T_\alpha\frac{\partial^2 P_\alpha}{\partial\gamma^2}\right] \tag{14}$$

8 Numerical Implementation Algorithm

1. Assemble $\mathcal{H}$ (Modulus + Stress terms), $\boldsymbol{\Xi}$, and $\mathcal{K}$.
2. Solve for all ω_m^2 and $\mathbf{u}^m$ using `Eigen::SelfAdjointEigenSolver`.
3. For each mode m:
   - If $\omega_m^2 > 10^{-14}$: Add $(\boldsymbol{\Xi}\cdot\mathbf{u}^m)^2/\omega_m^2$ to the correction.
   - If $\omega_m^2 < 10^{-14}$ AND $|\boldsymbol{\Xi}\cdot\mathbf{u}^m| > 10^{-10}$: Return $g = 0$ (floppy state).
4. Result: $g = (\mathcal{K} - \text{Correction})/(4N_c)$.

9 Physical Intuition & Illustrations

9.1 Edge Rotation and Stretching ($\frac{\partial l}{\partial\gamma}$)



$$\frac{\partial l_{ij}}{\partial\gamma} = \frac{\Delta x_{ij}\Delta y_{ij}}{l_{ij}} = L_y\cos\theta \text{ (for y-crossing)}$$

9.2 Lees-Edwards Unwrapping & Area Derivative

This diagram illustrates why $\frac{\partial A}{\partial \gamma} \neq 0$ under LEBC. As the box shears, the "unwrapped" ghost cell is shifted horizontally by γL_y. For a cell crossing the boundary, this shift changes the shoelace-calculated area if only internal coordinates are tracked.

